ICU Admission, Management, and Safety Guidelines

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XYZ HOSPITALS

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1.1 General Principles

1.1.1 ICU Admission Criteria

- Actual or imminent failure of ≥1 vital organ system:
- Circulatory: MAP <65 mmHg, need for vasopressors, lactate ≥4 mmol/L.
- Respiratory: PaO_2/FiO_2 <300, SpO_2 <90% despite oxygen, need for NIV or invasive ventilation.
- Renal: Rising creatinine, oliguria/anuria.
- Neurologic: Coma, agitation with risk to self/lines, uncontrolled seizures.
- Metabolic: Severe acidosis, DKA/HHS with hemodynamic compromise.
- Hematologic: Massive GI bleed, coagulopathy needing close monitoring.

1.1.2 Disposition Logic

- Full ICU: Shock, ARDS, refractory hypoxemia, mechanical ventilation, high-dose vasopressors, high bleeding risk, delirium compromising care.
- Step-down / HDU: Stable MAP without vasopressors, moderate oxygen requirement, improving organ function.
- Ward: Stable vitals, resolving pathology, no organ support beyond low-flow oxygen and routine IV meds.

1.1.3 Length of Stay (LOS) Estimation

- Complex shock / severe ARDS / post-MI: often 5–10 days, sometimes more.
- Moderate ARDS, COPD exacerbation, DKA/HHS, UGIB: typically 2–5 days.
- Stable LGIB, delirium, post-op neurosurgery: ICU 1–3 days, then step-down.

1.2 Septic and Distributive Shock Archetypes

1.2.1 Septic Shock from Pneumonia with Early ARDS

1.2.1.1 Syndrome profile

- Hypotension (MAP ~58) despite fluids.
- Tachycardia, tachypnea, high fever.
- WBC markedly elevated, lactate >5 mmol/L.
- Creatinine and bilirubin elevated, platelets reduced → early multiorgan dysfunction.
- PaO_2/FiO_2 ≈120 → moderate ARDS, SpO_2 <90% on baseline support.
- Already on mechanical ventilation with low tidal volume, high PEEP; norepinephrine + vasopressin; invasive lines.

1.2.1.2 ICU admission decision

- Definite ICU case: septic shock + ARDS + multiorgan failure + vasopressors + ventilation.
- Requires high-intensity, minute-to-minute management.

1.2.1.3 Immediate management priorities

1.2.1.4 1. Sepsis Bundle

- Continue broad-spectrum IV antibiotics (reassess daily, de-escalate once cultures return).
- Maintain aggressive, but controlled, fluid resuscitation guided by dynamic indices (pulse pressure variation, echocardiography).
- Titrate norepinephrine as first-line vasopressor; add vasopressin as per protocol to reduce norepinephrine dose.
- Target MAP ≥65 mmHg, lactate clearance (repeat every 2–4 hours).

1.2.1.5 2. ARDS Management

- Low tidal volume ventilation (4–6 mL/kg predicted body weight).
- Limit plateau pressures (<30 cmH_2O) and driving pressures.
- Use PEEP based on ARDS tables and response.
- Conservative fluids once shock is controlled.

1.2.1.6 3. Organ Dysfunction Monitoring

- Serial SOFA-type assessments using creatinine, bilirubin, platelets, PF ratio.
- Daily evaluation for renal replacement therapy if creatinine rises or urine output drops.
- Nutrition via early enteral feeds.

1.2.1.7 Expected ICU LOS

- Typically 7–10 days, longer if ventilator weaning is difficult or if secondary infections occur.

1.2.1.8 De-escalation and step-down criteria

- Stable MAP ≥65 mmHg without vasopressors for 24–48 h.
- PaO_2/FiO_2 improving (>200), lower PEEP, progressing toward spontaneous-breathing trials.
- Creatinine and bilirubin plateauing or improving; platelets recovering.
- Antibiotic course clearly defined; source control achieved.
- At that point:
- Remain in ICU if still ventilated but nearing extubation.
- Step-down after successful extubation and no vasopressors.

1.2.2 Severe Sepsis from Urinary Tract Infection (No Shock)

1.2.2.1 Syndrome profile

- Fever, tachycardia, tachypnea; MAP in low-normal range (~70–72).
- WBC elevated, lactate ~2.4 (mildly high).
- Creatinine slightly elevated, indicating early renal dysfunction.
- PaO_2/FiO_2 ≈320 → no ARDS; SpO_2 ≥95% on nasal cannula.
- On IV fluids, IV antibiotics, supplemental O_2, and urinary catheter.

1.2.2.2 ICU vs step-down

- No vasopressor requirement, MAP maintained, no severe hypoxemia.
- Suitable for HDU / step-down unit rather than high-intensity ICU unless:
- Rapid deterioration in MAP or lactate.
- Worsening mental status.
- Oliguria or rising creatinine.

1.2.2.3 Management priorities

- Ensure early and adequate antibiotics, tailored to urinary source, with culture-driven de-escalation.
- Maintain Euvolemia: initial IV fluids, then balance to avoid overload.
- Monitor renal function, urine output hourly initially, then q4h.
- Lactate trending: aim for normalization (<2 mmol/L).

1.2.2.4 Expected LOS

- 2–4 days in hospital.
- ICU/HDU stay <48 hours unless complications.

1.2.2.5 Step-down/discharge

- Transfer to ward once:
- MAP stable without fluid boluses.
- Lactate normal, creatinine stable.
- Pain and sepsis symptoms controlled, oral intake possible.
- Discharge home when afebrile for 24–48 h, ambulating, and able to continue oral antibiotics.

1.2.3 Anaphylactic Shock after IV Antibiotic

1.2.3.1 Syndrome profile

- Sudden hypotension (MAP ~50), tachycardia >130.
- SpO_2 mid-80s despite oxygen, PaO_2/FiO_2 ~200.
- Bronchospasm, airway compromise risk.
- Lactate mildly–moderately elevated due to hypoperfusion.
- Primary therapy: epinephrine IM then IV infusion, large-volume fluids, antihistamines, steroids, bronchodilators.

1.2.3.2 ICU admission

- Always ICU in the acute phase: risk of airway collapse, recurrence of shock, biphasic reactions.

1.2.3.3 Requires immediate access to:

- Advanced airway management.
    - Titrated epinephrine infusion and hemodynamic monitoring.

- Immediate management
    - Airway and Breathing
    - High-flow oxygen; prepare for intubation if stridor, altered mental status or fatigue.
    - Bronchodilators (nebulized beta-agonists, anticholinergics).
- Circulation
- Epinephrine: cornerstone; IM early, IV infusion in ICU.
- Rapid IV fluids (crystalloids).
- Monitor MAP, heart rate, urine output.

1.2.3.4 Adjuncts

- Antihistamines, corticosteroids to reduce late-phase response.
- Consider prophylaxis against biphasic anaphylaxis (observation ≥24 h for severe cases).

1.2.3.5 Expected LOS

- Many stabilize within 24–48 hours.
- ICU stay often 1–2 days, then ward or home with epinephrine auto-injector and allergy consultation.

1.2.3.6 Discharge criteria

- Hemodynamic stability without vasopressors.
- No respiratory compromise off supplemental O_2.
- No recurrence of symptoms for 24 hours.
- Clear trigger identified and documented in medical record.

1.3 Cardiogenic and Neurogenic Shock Archetypes

1.3.1 Cardiogenic Shock after Anterior STEMI

1.3.1.1 Syndrome profile

- MAP ~55, lactate ~4, moderate hypoxemia (PaO_2/FiO_2 ~260).
- Elevated creatinine → renal hypoperfusion.
- On mechanical ventilation, inotropes (dobutamine) and vasopressor (norepinephrine), possible IABP.

1.3.1.2 ICU admission

- Mandatory ICU: post-MI cardiogenic shock has high mortality and requires continuous hemodynamic support and close arrhythmia monitoring.

1.3.1.3 Management priorities

- Circulatory Support
  - Inotrope (dobutamine) to augment cardiac output.
  - Norepinephrine to maintain MAP ≥65 if needed.
  - Consider mechanical circulatory support (IABP, Impella, ECMO) based on institutional protocols.
- Post-MI Care
  - Dual antiplatelet therapy as per cardiology.
  - Anticoagulation, statins, beta-blockers as tolerated.
  - Strict arrhythmia surveillance; correct electrolytes.
- Fluid Strategy
  - Very cautious IV fluids; avoid pulmonary edema.
  - Diuretic infusions to offload congestion if lungs overloaded.

1.3.1.4 Expected LOS

- Often 5–7 days in ICU if stabilized, longer if ongoing cardiogenic shock or advanced support used.
- Step-down/discharge
- Step-down when:
  - Off mechanical circulatory support and high-dose inotropes.
  - Stable rhythm, MAP, and lactate.
  - Diuretics manageable via peripheral lines, and oxygen requirement is low.
  - Later discharge from hospital after cardiac rehab planning.

1.3.2 Neurogenic Shock after High Cervical Spinal Cord Injury

1.3.2.1 Syndrome profile

- Hypotension (MAP low-50s) with relative bradycardia (~50s).
- Lactate normal; WBC normal; PaO_2/FiO_2 normal.
- Peripheral vasodilation; warm extremities.
- On vasopressors (norepinephrine/phenylephrine) for spinal cord perfusion.

1.3.2.2 ICU admission

- Yes, ICU for:
  - Maintenance of high MAP targets (often ≥85–90) to optimize spinal cord perfusion.
  - Continuous hemodynamic and neurological monitoring.

1.3.2.3 Management priorities

- Hemodynamics
  - Titrate vasopressors to target MAP based on neuro-protective guidelines.
  - Avoid excessive fluids once euvolemic.
- Airway/Spine Protection
  - Cervical collar; maintain spine precautions.
  - Early involvement of neurosurgery and trauma teams.
- Secondary Complications
  - Prevent pressure sores, DVT, and pneumonia.
  - Early physio and occupational therapy planning.

1.3.2.4 Expected LOS

- ICU 3–7 days, often followed by prolonged rehab stay.

1.3.2.5 De-escalation

- Step-down when:
  - MAP stable at target with minimal vasopressor support.
  - Autonomic instability reduced.
  - Safe respiratory status and spine stabilized surgically or conservatively.

1.4 Respiratory Failure, ARDS and Obstructive Disease

1.4.1.1 Moderate ARDS Secondary to Pancreatitis

1.4.1.2 Syndrome profile

- Tachycardic, tachypneic, febrile, MAP ~70 (borderline).
- WBC elevated, lactate mildly up; PaO_2/FiO_2 ~160 → moderate ARDS.
- On mechanical ventilation, low tidal volume, PEEP ~10, sedation/analgesia.

1.4.1.3 ICU admission

- Yes: ARDS requiring invasive ventilation.

1.4.1.4 Management

- ARDSNet low tidal-volume strategy (4–6 mL/kg).
- Moderate PEEP (e.g., 8–12 cmH_2O) guided by oxygenation and compliance.
- Conservative fluid management, especially once pancreatitis stabilized.
- Pain and sedation tailored to allow ventilator synchrony.

1.4.1.5 LOS

- Typically 4–7 days in ICU, depending on course of pancreatitis and respiratory recovery.

1.4.1.6 Severe ARDS due to Influenza Pneumonia, Refractory Hypoxemia

1.4.1.7 Syndrome profile

- Fever, tachycardia, MAP ~65 on vasopressor support.
- PaO_2/FiO_2 ≈80 → severe ARDS.
- SpO_2 mid-80s even with high FiO_2, high PEEP; requires deep sedation, paralysis and prone positioning.

1.4.1.8 ICU admission

- Definite ICU with advanced respiratory support; potential ECMO candidate.

1.4.1.9 Management priorities

- Low tidal-volume ventilation, high PEEP with careful hemodynamic assessment.
- Scheduled prone positioning (e.g., ≥16 hours/day).
- Neuromuscular blockade during the most severe phases.
- Early discussions with ECMO center if refractory to maximal conventional therapy.

LOS

- Often 7–14+ days in ICU; prolonged ventilation is common.

1.4.1.10 COPD Exacerbation with Acute Hypercapnic Respiratory Failure

1.4.1.11 Syndrome profile

- Tachypneic (RR ~30), hypoxemic (SpO_2 ~88%), moderate tachycardia.
- PaO_2/FiO_2 ~260, but underlying hypercapnia expected.
- Managed with NIV (BiPAP), bronchodilators, steroids, antibiotics, targeted O_2 (88–92%).

1.4.1.12 ICU admission

- ICU or respiratory HDU depending on staffing:
- Indicated when NIV required plus significant risk of intubation.

1.4.1.13 Management

- Close monitoring of pH and $PaCO_2$ via ABGs.
- Adjust BiPAP settings to reduce work of breathing.
- Avoid over-oxygenation to prevent CO_2 retention.
- Treat triggers (infection, cardiac issues).

1.4.1.14 LOS

- ICU 2–4 days, then ward once stable on low-flow O_2 or room air.

1.4.1.15 # 4.4 Status Asthmaticus with Impending Respiratory Failure

1.4.1.16 Syndrome profile

- Extreme tachypnea, tachycardia, accessory muscle use.
- SpO_2 high 80s/low 90s; PaO_2/FiO_2 ~230.
- On continuous nebulized bronchodilators, IV steroids, magnesium, NIV trial; epinephrine PRN.

1.4.1.17 ICU admission

- Yes, due to risk of sudden decompensation and need for intubation.

1.4.1.18 Management

- Aggressive bronchodilation (continuous nebs).
- Steroids and magnesium sulphate.
- NIV as bridge; watch for:

- Rising CO_2 with clinical fatigue and “silent chest” → intubate early.
- If intubated:
- Permissive hypercapnia.
- Prolonged expiratory time to avoid air-trapping.

1.4.1.19 LOS

- ICU 2–5 days depending on response and need for intubation.

1.4.1.20 Metabolic Emergencies: DKA and HHS

1.4.1.21 Diabetic Ketoacidosis with Mild Shock

1.4.1.22 Syndrome profile

- Tachycardia, borderline MAP (~68), Kussmaul or rapid respirations.
- Lactate mildly elevated; creatinine slightly high.
- Managed with IV fluids, insulin infusion, potassium replacement; frequent BMP.

1.4.1.23 ICU admission

- Recommended when:
- Hemodynamic compromise (borderline MAP).
- High anion-gap acidosis.
- Insulin infusion requiring hourly titration and strict monitoring.

1.4.1.24 Management

- Standard DKA protocol:
- Initial bolus of isotonic saline then adjusted fluids.
- Insulin infusion after potassium levels confirmed.
- Avoid overly rapid shifts in osmolality.
- Hourly glucose checks; 2–4 hourly electrolytes and gases.

1.4.1.25 LOS

- ICU 1–2 days until anion gap closed and hemodynamics stable.
- Total hospital stay 3–5 days.

1.4.1.26 Hyperosmolar Hyperglycemic State with Pre-renal AKI

1.4.1.27 Syndrome profile

- Older adult, tachycardia but reasonably preserved MAP (~74).
- Lactate normal; creatinine ~2 indicating AKI from volume depletion.
- No major respiratory compromise.

1.4.1.28 ICU vs step-down

- Can be managed in ICU or a well-staffed HDU, especially early in resuscitation because:
- Large-volume fluid resuscitation.
- Gradual osmolar correction requires close sodium/glucose monitoring.

1.4.1.29 Management

- Aggressive but controlled fluid replacement over 24 h.
- Low-dose insulin infusion.
- Electrolyte and osmolarity tracking to avoid cerebral edema.
- Foley for strict I/O.

1.4.1.30 LOS

- ICU 1–3 days, then ward until metabolic normalization and functional recovery.

1.4.1.31 Gastrointestinal Bleeding

1.4.1.32 Upper GI Bleed with Hemodynamic Compromise

1.4.1.33 Syndrome profile

- Tachycardia ~125, MAP ~60, lactate ~3.3.
- Risk factors for liver disease or anticoagulation may be present.
- On IV fluids, packed RBC transfusions, IV PPI, octreotide, NG tube; endoscopy planned.

1.4.1.34 ICU admission

- Yes:
- Active bleeding with hypotension and elevated lactate.
- Potential need for massive transfusion and airway protection.

1.4.1.35 Management

- Initial resuscitation: two large-bore IVs; balanced transfusion as per threshold (e.g., Hb <7–8 g/dL, individualized).
- Proton pump inhibitor infusion; octreotide if variceal bleed suspected.
- Correct coagulopathy (vitamin K, PCC).
- Plan for urgent endoscopy (ideally within 24 hours, earlier for variceal bleed).

1.4.1.36 LOS

- ICU 2–4 days if bleeding controlled; longer if rebleeding or multi-organ impact.

1.4.1.37 Lower GI Bleed with Relative Hemodynamic Stability

1.4.1.38 Syndrome profile

- Mild tachycardia, MAP ~78, lactate normal.
- On moderate IV fluids, type and screen, colonoscopy planned.

1.4.1.39 ICU vs ward

- Typically does not require ICU if:
- No ongoing hemodynamic instability.
- Lactate normal and no rapid Hb drop.
- Could be observed in step-down if high-risk comorbidities.

1.4.1.40 LOS

- Hospital stay 2–4 days, rarely ICU.

1.4.1.41 Neuro-Behavioral and Post-Operative ICU Archetypes

1.4.1.42 Hyperactive ICU Delirium in Recovering Sepsis

1.4.1.43 Syndrome profile

- Vitals stable; labs improving; PF ratio good.
- Main issue: hyperactive delirium – agitation, pulling lines, sleep–wake reversal.
- Managed with minimal sedation, low-dose antipsychotics, non-pharmacologic bundle.

1.4.1.44 ICU admission

- Already in ICU; the key decision is when to step-down.
- May remain temporarily if:
- Risk of self-harm or device removal is high.

- Requires close nursing ratio for safety.

1.4.1.45 Management

- Use CAM-ICU or similar tool for diagnosis.
- Optimize sleep hygiene, reduce nighttime noise/light.
- Early mobilization, reorientation, family presence.
- Judicious antipsychotic use, avoiding oversedation.

1.4.1.46 LOS

- Additional 1–3 days in ICU until behavior more predictable and safe for ward.

1.4.1.47 # 7.2 Post-operative Neurosurgical Patient with BP Control and Light Sedation

1.4.1.48 Syndrome profile

- Hemodynamically stable (MAP ~95), HR normal, PF ratio excellent.
- On mechanical ventilation with spontaneous breathing trials planned.
- Nicardipine for BP control; light sedation targeting RASS -1 to 0; ICP monitoring in place.

1.4.1.49 ICU admission

- Yes, due to:

- Need for tight BP and ICP control.
- Risk of acute neurologic deterioration post-op.

1.4.1.50 Management

- Maintain MAP and systolic BP within neurosurgical targets.
- Monitor ICP and CPP as per protocol.
- Daily sedation interruption and spontaneous breathing trials to plan extubation.
- Early CT brain or neuroimaging if any neurological change.

1.4.1.51 LOS

- Usually 2–4 days in ICU, then transfer to neuro step-down once:
- Extubated.
- BP controlled with oral meds.
- ICP monitoring no longer required.

1.4.1.52 8. High-Level Disposition Summary by Archetype

1.4.1.53 Must-start-in-ICU (full support):

- Septic shock with ARDS.
- Cardiogenic shock post-MI.
- Neurogenic shock with MAP targets.
- Severe ARDS with refractory hypoxemia.
- Anaphylactic shock.
- DKA with shock features (early phase).
- Upper GI bleed with hemodynamic compromise.
- Post-op neurosurgical patient needing ICP/BP control.

1.4.1.54 ICU or HDU (depending on resources/acuity):

- Severe sepsis without shock.
- HHS with AKI.
- COPD exacerbation on NIV.
- Status asthmaticus with impending failure.
- Delirium with safety concerns.

1.4.1.55 Usually non-ICU:

- Lower GI bleed with stable hemodynamics (unless comorbid risk).

1.5 ICU Admission Criteria

- Actual or imminent failure of one or more organ systems requiring continuous monitoring and/or advanced support.
- Circulatory compromise: mean arterial pressure (MAP) <65 mmHg, need for vasopressors or inotropes, or lactate ≥4 mmol/L despite adequate fluid resuscitation.
- Respiratory failure: severe hypoxemia (e.g., SpO_2 <90% on supplemental oxygen or PaO_2/FiO_2 <300), increased work of breathing, need for non-invasive ventilation (NIV) or invasive mechanical ventilation.
- Neurologic compromise: coma, persistent seizures, severe agitation or delirium that threatens patient safety or interferes with essential therapy.
- Metabolic emergencies: DKA or HHS with hemodynamic instability, severe acidosis, or rapidly evolving electrolyte disturbances requiring continuous adjustment.
- Major bleeding: active upper GI bleed, post-operative hemorrhage, or other causes with hypotension, rising lactate, or need for massive transfusion.

- Post-operative high-risk surgeries: neurosurgical, major cardiovascular, or high-risk abdominal procedures where close hemodynamic and neurologic monitoring is needed.

1.6 Situations Where ICU Admission May Not Be Necessary

- Patients who are hemodynamically stable (MAP ≥65 mmHg without vasopressors) and have no severe hypoxemia or respiratory distress.
- Sepsis without shock, with mild organ dysfunction that can be safely monitored in a high-dependency unit (HDU) or step-down ward with frequent reassessment.
- Non-massive lower GI bleed with stable hemoglobin, normal lactate, and no evidence of ongoing hemodynamic compromise.
- Metabolic abnormalities that can be managed with scheduled monitoring and without continuous infusions or invasive support.

1.7 ICU Discharge (Step-down/Transfer) Criteria

- Hemodynamic stability: MAP consistently ≥65 mmHg without vasopressors for at least 24 hours, no requirement for rapid fluid boluses, and stable heart rate without malignant arrhythmias.
- Respiratory stability: no need for invasive ventilatory support, or transitioned to low-level oxygen therapy (e.g., nasal cannula) with SpO_2 ≥92% (or individualized target for COPD) at rest.
- Improving or stable organ function: creatinine, bilirubin, platelets, and lactate trending toward normal or at new baseline; no rapidly evolving organ failure.
- Neurologic stability: no uncontrolled seizures, agitation manageable without continuous sedatives or physical restraints, level of consciousness appropriate for step-down care.
- No high-risk procedures or infusions requiring ICU-level nurse-to-patient ratios (e.g., high-dose vasopressors, complex antiarrhythmic infusions, or ICP monitoring).
- Clear plan for ongoing care, including oral or intermittent IV medications, ward-appropriate monitoring, and follow-up imaging or procedures.
- 2. Syndrome-Based ICU Management (Overview)
- Septic Shock with Pneumonia and Early ARDS: Characterized by hypotension despite fluids, high lactate, multiorgan dysfunction, and hypoxemic respiratory failure. Requires invasive ventilation with low tidal-volume strategy, vasopressors (often norepinephrine plus vasopressin), lactate-guided resuscitation, and early broad-spectrum antibiotics. Typical ICU LOS: 7–10 days, depending on response and source control.
- Severe Sepsis from Urinary Tract Infection (No Shock): Features fever, leukocytosis, mildly elevated lactate, and early renal dysfunction without vasopressor requirement. Often suitable for HDU or step-down rather than full ICU, unless deterioration occurs. Core management includes IV fluids, antibiotics, and close monitoring of renal function and lactate.
- Cardiogenic Shock after Myocardial Infarction: Marked by low MAP, elevated lactate, signs of pulmonary congestion, and reduced cardiac output. Requires inotropes (e.g., dobutamine), cautious vasopressors, possible mechanical circulatory support, and post-MI care. Fluid resuscitation must be judicious to avoid fluid overload.
- Neurogenic Shock after High Cervical Spinal Cord Injury: Characterized by hypotension with relative bradycardia, warm extremities, and normal lactate. Management centers on vasopressor support to maintain neuroprotective MAP targets, spinal stabilization, and prevention of complications such as pressure injuries and DVT.
- Anaphylactic Shock after Drug Exposure: Rapid-onset hypotension, bronchospasm, and hypoxemia following exposure to an allergen. Epinephrine is the primary therapy, supported by IV fluids, antihistamines,

corticosteroids, and bronchodilators. ICU admission is required initially, with typical LOS of 1–2 days if recovery is uncomplicated.

- Moderate and Severe ARDS: Defined by impaired oxygenation (PaO_2/FiO_2 ≤200 or ≤100) with bilateral pulmonary infiltrates not fully explained by cardiac failure. Managed with low tidal-volume ventilation, appropriate PEEP, conservative fluids, and in severe cases, prone positioning, neuromuscular blockade, and consideration of ECMO.
- Acute Exacerbation of COPD or Status Asthmaticus: Obstructive airway diseases with acute respiratory failure managed with bronchodilators, steroids, and non-invasive ventilation when possible. ICU-level care is indicated when there is impending respiratory failure, need for intubation, or close monitoring of gas exchange and work of breathing.
- Metabolic Emergencies: DKA and HHS: High-glucose states associated with dehydration, electrolyte disturbances, and acidosis (particularly in DKA). ICU care is needed when there is hemodynamic compromise, severe acidosis, or requirement for insulin infusion and frequent laboratory monitoring.
- GI Bleeding with Hemodynamic Compromise: Upper GI bleeding can cause shock and elevated lactate, requiring rapid resuscitation, blood transfusion, PPI and adjunctive therapies, and urgent endoscopy. ICU admission is warranted when there is hypotension, ongoing bleeding, or high transfusion requirements.
- Delirium and Post-operative Neurosurgical Care: Patients recovering from sepsis or neurosurgery may need ICU-level monitoring for delirium, ICP management, tight blood pressure control, and safe progression toward extubation. Delirium management prioritizes non-pharmacologic measures and cautious use of antipsychotics.

1.8 Infection Prevention and Disinfection in the ICU

- Preventing healthcare-associated infections (HAIs) is fundamental to ICU safety. While institutional policies vary, the following general principles are widely applicable and can be encoded into workflow checklists or RAG-based decision support.

1.9 Hand Hygiene

- Perform hand hygiene before touching a patient, before aseptic procedures, after exposure to bodily fluids, after touching a patient, and after touching patient surroundings.
- Use alcohol-based hand rub for most indications; use soap and water when hands are visibly soiled or when caring for patients with certain pathogens (e.g., C. difficile).
- Ensure that sinks, dispensers, and supplies for hand hygiene are conveniently located throughout the ICU.

1.10 Personal Protective Equipment (PPE)

- Use gloves, gowns, masks, and eye protection based on the anticipated exposure (standard, contact, droplet, or airborne precautions).
- Don PPE before entering the room when isolation signs indicate enhanced precautions, and doff PPE carefully to avoid self-contamination.
- Ensure proper fit and seal for respirators (e.g., N95) when caring for patients with airborne infectious diseases.

1.11 Environmental Cleaning and Disinfection

- High-touch surfaces (bed rails, monitors, IV pumps, bedside tables) should be cleaned and disinfected at least once per shift and after visible contamination.
- Terminal cleaning should be performed after patient discharge, using disinfectants compatible with institutional policies and equipment manufacturers' recommendations.
- Reusable equipment (e.g., stethoscopes, ultrasound probes) should be cleaned between patients, following device-specific protocols.

1.12 Device-Associated Infection Prevention

- Central lines: use full barrier precautions during insertion, chlorhexidine skin prep, and strict maintenance bundles; remove lines as soon as they are no longer essential.
- Urinary catheters: insert only when necessary, maintain aseptic technique, avoid routine irrigation, and reassess daily for removal.
- Endotracheal tubes and ventilators: elevate head of bed (30–45 degrees), perform oral care, use sedation and weaning protocols, and minimize the duration of mechanical ventilation.

1.13 ICU Safety Protocols

1.13.1 Patient Identification and Communication

- Use at least two patient identifiers (e.g., name and medical record number) before any procedure, medication administration, or specimen collection.
- Implement structured communication tools such as SBAR (Situation, Background, Assessment, Recommendation) for handovers and critical updates.
- Document all changes in treatment plans and ensure that verbal orders are read back and confirmed.

1.13.2 Medication Safety

- Standardize concentration and labeling of high-alert medications (e.g., vasopressors, anticoagulants, insulin).
- Use independent double-checks for high-risk infusions, especially when starting or changing doses.
- Employ smart pumps and dose-error reduction systems when available, and avoid look-alike/sound-alike medication confusion through clear labeling.

1.13.3 Line, Tube, and Device Safety

- Perform daily assessments of all lines, catheters, and drains to determine ongoing necessity and remove non-essential devices promptly.
- Secure all tubes and lines to minimize accidental dislodgement; label them clearly to avoid misconnections.
- Ensure that alarms on monitors and infusion pumps are active, audible, and appropriately configured for the patient's clinical status.

1.13.4 Falls, Restraints, and Delirium Prevention

- Assess fall risk routinely and employ environmental and staffing strategies (e.g., bed rails, frequent rounding) to reduce falls.
- Use physical restraints only as a last resort and for the shortest duration possible, adhering to institutional policies and frequent reassessment.
- Implement a delirium prevention bundle including sleep hygiene, early mobilization, minimizing unnecessary sedatives, and promoting sensory aids (glasses, hearing aids).

1.14 Length of Stay and Disposition Framework

- Length of ICU stay in these synthetic archetypes is an estimate intended for simulation and planning. Actual ICU LOS depends on patient response, complications, and local resource availability.
- Complex shock or severe ARDS (e.g., septic shock with multiorgan failure, cardiogenic shock, refractory ARDS): often 5–14 days, sometimes longer.
- Moderate ARDS, COPD/asthma exacerbations requiring ventilation or NIV, DKA/HHS with hemodynamic compromise, and upper GI bleeds: typically 2–7 days.
- Delirium-focused cases or post-operative neurosurgical monitoring without major complications: usually 1–4 days in ICU, followed by step-down or ward care.

- Stable lower GI bleeds or improving metabolic/sepsis syndromes with no organ support: managed largely in step-down/ward settings with short or no ICU stay.